

Quiz 01

Emory University

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Instructions

This quiz covers material from **Lectures 01–07**. Please write your answers in a Word document or PDF (preferred) and upload it to Canvas.

Guidelines:

- You are **encouraged** to use ChatGPT, Claude, Gemini, or other LLMs to help you think through these questions
 - However, you must **write your answers in your own words** and demonstrate genuine understanding of the concepts
 - Please **disclose which AI tool you used** (if any)
 - Your answers should be thoughtful and show critical thinking, not just copy-pasted LLM outputs
 - If applicable, please **attach screenshots** of your interactions with AI tools
 - Submit your file via **Canvas** as a PDF (preferred) or Word document
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Question 1: The “Fourth Word” Mistake

Imagine that you ask an AI model:

“What is the fourth word in this sentence?”

The model responds confidently:

“The fourth word in this sentence is ‘word.’”

This is wrong. The fourth word is “fourth.”

Your task:

1. Explain why the model made this mistake. Use the definition from Lecture 01 (Slide 29) and the specific term for how AI processes text.
2. What is the specific term for how AI processes text?
3. Why would asking the AI “Why did you get this wrong?” likely produce a vague philosophical answer rather than a mechanical explanation?

Question 2: “Dog Bites Man” vs “Man Bites Dog”

You input two sentences into a model:

- Sentence A: “*Dog bites man*”
- Sentence B: “*Man bites dog*”

A student claims:

“The model uses an embedding vector for ‘Man,’ ‘Dog,’ and ‘Bites.’ Since both sentences contain the same words, the sum of the vectors is identical. Therefore, the model cannot tell who did the biting.”

Your task:

1. Is this student correct? (Yes or No)
2. Are word vectors alone (without any additions) sufficient to distinguish the meaning of these two sentences?
3. What specific component is added to embeddings to ensure the model knows the order of words?

Question 3: The Medical Coding Problem

You are building a Medical Coding Assistant that converts doctors’ notes into billing codes (ICD-10). Accuracy is mandatory — the model must output the correct code every time.

You review the inference settings:

- **Temperature:** 0.9
- **Top-P:** 0.95

A colleague says: “*High temperature ensures the model thinks harder and is more creative in finding the correct solution.*”

Your task:

1. Is your colleague’s logic correct? (Yes or No)
2. What does a high Temperature setting do to the probability distribution of the next token?
3. What specific failure mode would this cause for medical billing codes?
4. Should Temperature be set High or Low for this task? Explain briefly.

Question 4: The Proxy Problem

A hospital wants to use AI to identify “high-performing nurses” for promotion.

- **Actual Goal:** Identify nurses who provide the best patient care.
- **Available Data (Proxy):** Number of patients seen per shift.

Your task:

1. Is “patients seen per shift” a good or bad proxy for “quality of care”? Explain why, referencing the concept of **measurement-reality gap** from the Proxy Problem lecture.
2. Describe ONE specific negative behaviour the AI might learn to reward.
3. Propose a better proxy, even if imperfect, that would more closely align with the actual goal.

Question 5: Semantic Search vs Keyword Search

A developer complains:

“Our search system is broken. I searched for ‘Canine nutrition,’ but the top result discussed ‘Dog food.’ The words ‘Canine’ and ‘nutrition’ don’t appear in the result. The Vector Database must be broken.”

Your task:

1. Is this developer correct that the system is broken? (Yes or No)
2. Using the concept of Embeddings and Vector Space, explain how the system matched these two phrases even though they share no keywords.